

EE 252 LAB DC-DC CONVERTER PROJECT REPORT

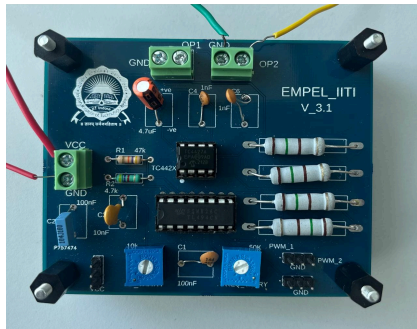
BATCH NO: B1-G7

ROLL NO: 240002022,240002030,240002036,240002042

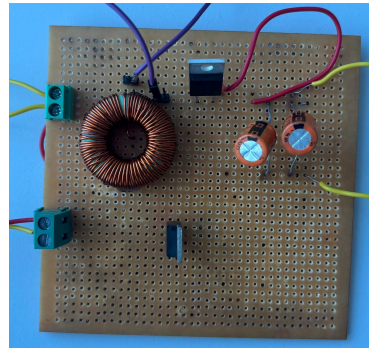
Buck - Boost Converter

a) Specifications: Input 15 V, Output 12-18 V, Switching frequency 12.5 kHz, Output current 1 A. Waveforms of diode current and inductor voltage in CCM.b) Reduce switching frequency to demonstrate DCM.

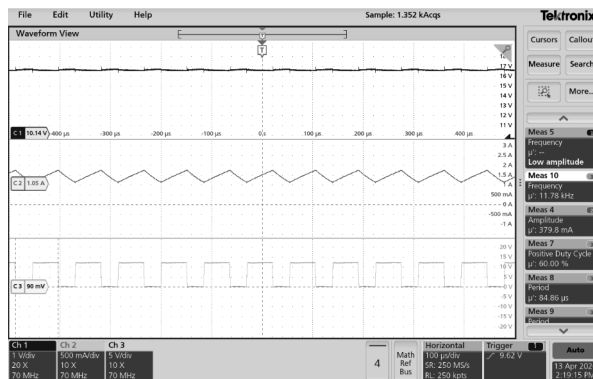
Soldered Gate Driver



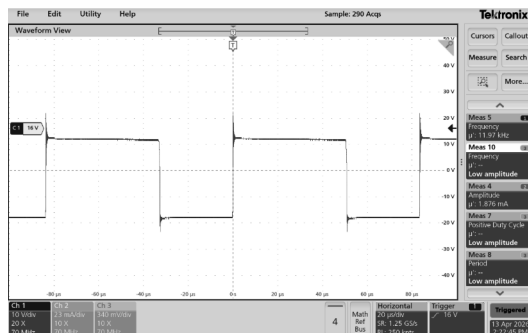
Soldered Buck-Boost Converter



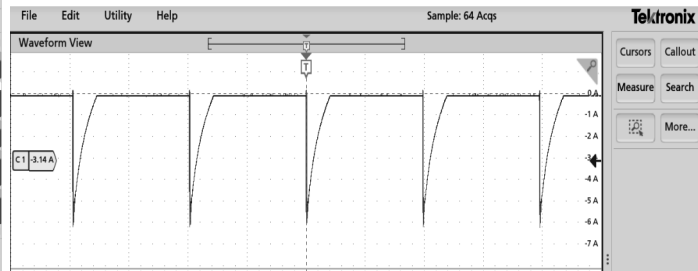
Waveforms



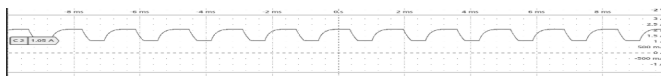
Output voltage, inductor current, and gate pulse in ccm



Output voltage, diode current, and gate pulse in ccm



Inductor Voltage in CCM



Diode current in DCM

Inductor current in DCM

[Matlab simulation](#)

[Calculations](#)